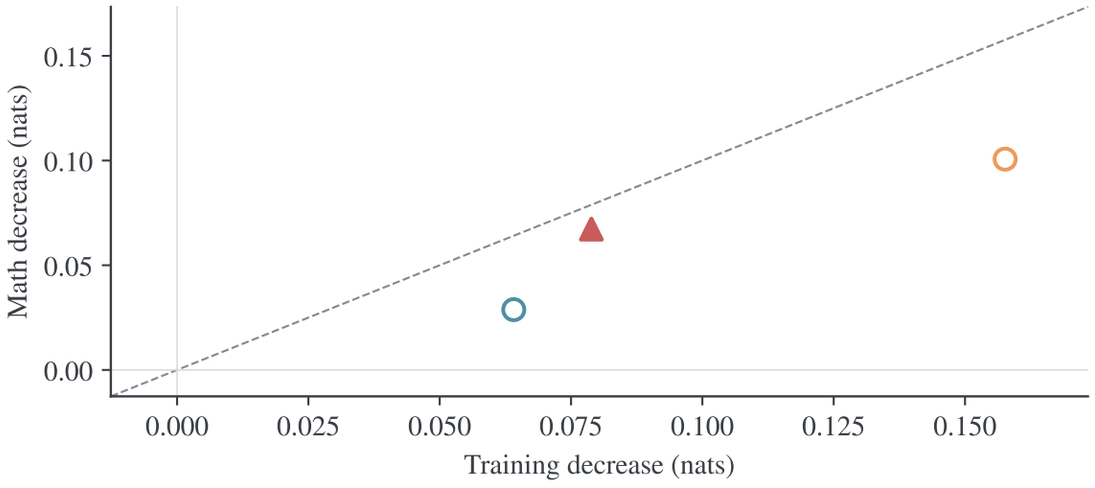
R1-7B | Training pool vs. held-out

Absolute RKL decrease (nats / H1024)

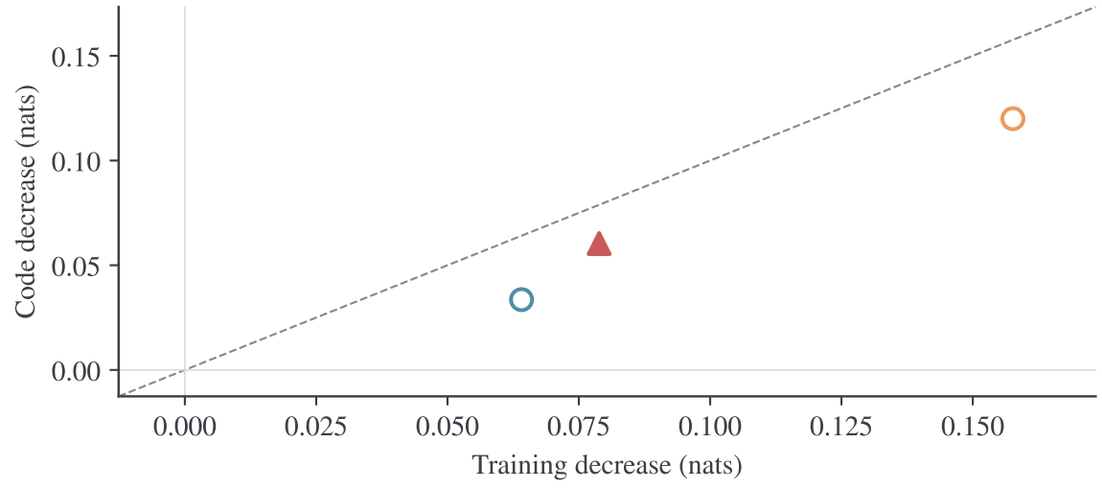
(a) Training vs. Math

n=3; r=0.91; rho=1.00



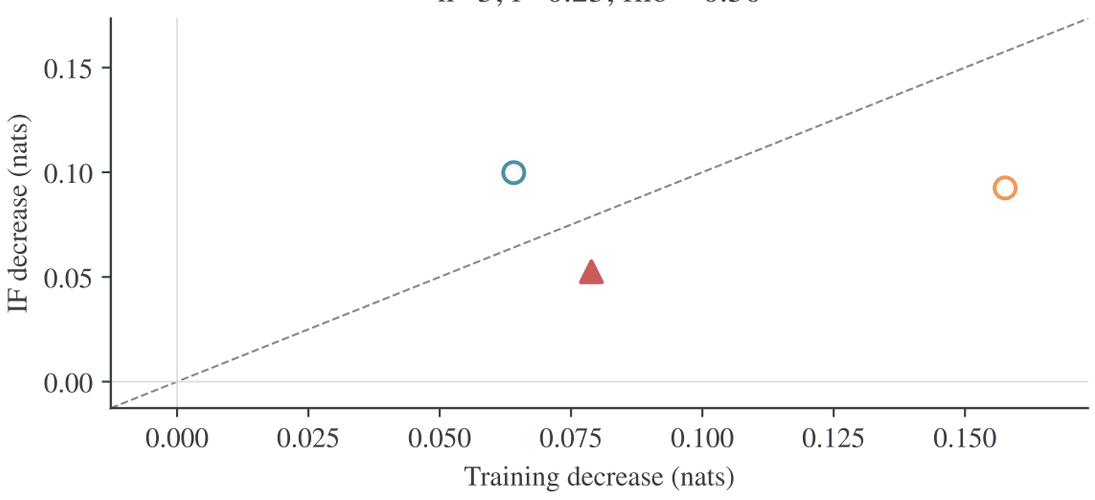
(b) Training vs. Code

n=3; r=0.99; rho=1.00



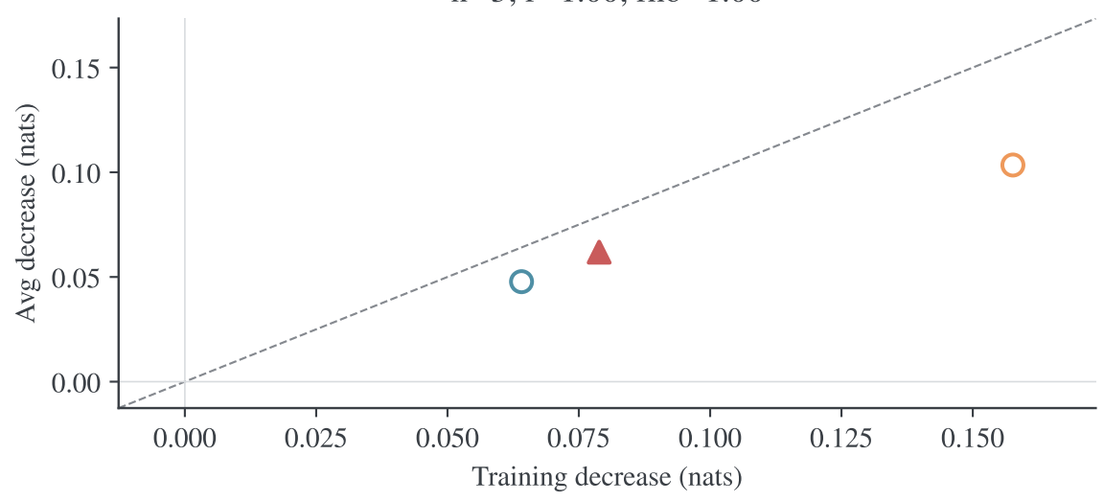
(c) Training vs. IF

n=3; r=0.23; rho=-0.50



(d) Training vs. Avg

n=3; r=1.00; rho=1.00



DeepSeek-R1-Distill-Qwen-1.5B [320]

JustRL-DeepSeek-1.5B [320]

Light-R1-7B-DS [64]

Equal decrease

One point = one run endpoint; triangle: 64 updates; circle: 320; filled: same size; open: cross size.

Fixed-H RKL; dashed: equality; correlations pool all displayed endpoints (mixed budgets/sizes).